Why Are There So Many Species of Flowering Plants?

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This essay is in part a reply to the companion one by Burger and in part supplementary to it. Since the topic is to a large extent one of taxonomy or systematics—and reflects also the evolutionary history of a class of organisms over 120 million years—data from systematics are essential, and paleontological data must be considered when they are available. (*Accepted for publication 30 April 1981*)

Burger (in the companion article) believes that "the most important innovation may have been pollination by insects, which allows outcrossing sexual reproduction in highly dispersed populations of relatively few individuals." I question this conclusion for the following reasons: First, it requires that during the early part of the Cretaceous period, insects already existed that were strong flyers at the same time strongly attracted to flowers. Although the fossil record of Cretaceous insects is meager, it suggests that the most efficient modern pollinators, particularly bees, had not evolved modern genera at that early period. Second, although insect pollination was most probably the basis of selection pressure that produced the hermaphroditic or "perfect" flower, which was characteristic of the earliest angiosperms, speciation based upon floral differences would require either adaptation to many different kinds of pollinators or a complex floral structure that could be modified in various ways to obtain different methods of pollination by similar vectors. I doubt that either of these conditions existed during the earliest stages of angiosperm evolution. Third, the ability of individuals to be cross-pollinated by vectors flying over long distances might hinder rather than promote speciation, since spatial isolation is the usual prelude to the evolution of those reproductive isolating mechanisms that cause species to become distinct from each other.

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I believe that one of the best ways to approach the problem is to capitalize on the enormous difference in size between categories having the same rank at all levels of the taxonomic hierarchy. In many families, monotypic genera exist side by side with giants that contain hundreds of species. Most orders contain a few large families and many smaller ones. Consequently, one may legitimately ask, What, if any, distinctive characteristics did the ancestors of the larger families' genera possess that enabled them to evolve so many species? And what was the basis of diversification of the largest families?

For my discussion, I posed three questions. The first considers eight of the largest angiosperm genera: *Senecio* (1200 species), *Astragalus* (1200), *Eucalyptus* (450), *Ficus* (600), *Erica* (450), *Cassia* (400), *Carex* (500), and *Panicum* (300). Why has speciation progressed so much farther in these genera than in their neighbors, most of which have fewer than 50 species and many of which are monotypic or contain only 2 to 5 species? Second, what has caused a few very large families to become rich and diverse in genera as well as in species, as are Asteraceae (1000 genera), Orchidaceae (800 genera), Fabaceae (sens. lat.) (700 genera), and Poaceae (700 genera)? Third question, since the mean number of species *per family* is no greater than it is in the most successful non-angiospermous vascular plants (e.g., Filicales or ferns), Why have angiosperms evolved and maintained some 250 different families, as compared to 3 of the class Lycopsida, 10 of Pteropsida, 1 each of Sphenopsida, Cycadales, and Ginkgoales, 3 of Gnetales, and 7 of Coniferales? Given this large number of families, a total number of 250,000 species might be expected to evolve even at rates no faster than those that prevail in other orders and classes of plants; the mean number of species per family in leptosporangiate ferns (8 families, 9000 species) is nearly the same as in angiosperms (250 families, 250,000 species). These three questions will be taken up in turn.

WHY DO LARGE GENERA EVOLVE?

Even a brief perusal of the eight very large genera mentioned above ascertains the fact that no single explanation fits all of them. They differ greatly from each other on every parameter. With respect to geographic distribution, *Eucalyptus* is almost entirely endemic to a single continent (Australia), *Ficus* is pantropical but almost absent in temperate regions, *Astragalus* and *Carex* are strictly temperate and arctic, and *Cassia* and *Panicum* have worldwide distributions in both tropical and temperate zones. With respect to age, none of these genera is known to have existed earlier than the Eocene epoch, 50 to 60 million years ago, but evidence from geographic distribution suggests that speciation in *Ficus* and *Cassia* may have begun during the Eocene (Raven and Axelrod 1974). The geographical evidence also leads me to believe that speciation in *Astragalus*, *Senecio*, and *Carex* took place chiefly during the Miocene and Pliocene epochs, i.e., the last 15 million years. With respect to growth habit, *Cassia* has evolved a great diversity, including trees, shrubs, perennial, and annual herbs, whereas *Eucalyptus*, *Ficus*, and *Carex* are no more diverse than are many smaller genera belonging to their families.

Speciation in *Ficus* is based chiefly upon the coevolution of this genus and the fig wasp *Blastophaga* with respect to a complex and highly specific method of crosspollination (Baker 1961, Ramirez 1970). In *Erica* also, floral diversity, associated with different pollinators and devices for crosspollination, plays an important role in the diversification of species. The same factor appears to have been important in differentiating the eight sections of *Cassia*. On the other hand, *Carex* and *Panicum* are both chiefly wind-pollinated and have highly stereotyped androecia, whereas floret structure in *Senecio* is likewise stereotyped as in the androecium of *Eucalyptus*. Diversity of floral structure at anthesis in *Astragalus* is no greater than in other herbaceous genera of Fabaceae.

The species of *Carex* and *Astragalus* owe their diversity chiefly to a great range of variation with respect to fruits and their accessory structures, which gave rise to a multiplicity of devices for seed dispersal. On the other hand, species of *Ficus*, *Eucalyptus*, and *Senecio* are much like each other in this respect.

With respect to chromosomes, the species of *Ficus* nearly all have the somatic number $2n = 26$, whereas those of *Eucalyptus* have $2n = 22$. In both genera, chromosomes are relatively small and differ little from one species to another in size and form. The same is true of *Erica* ($2n = 24$), although relatively few species of this genus have been counted (Fedoroff 1969). *Senecio, Panicum*, and *Cassia* contain a high proportion of polyploids. Aneuploidy is moderately well-developed in *Astragalus* species of the New World; polyploidy exists among Old World species. The genus *Carex* is well known for its extensive aneuploid series.

Evolution of the annual growth habit, associated with self-compatibility and autogamy, is often invoked as a cause of extensive speciation in angiosperms (Grant 1971). Of the eight genera under consideration, *Eucalyptus, Ficus, Erica*, and *Carex* contain no annual species at all. Among the other four, annuals are distinctly in the minority, having added not more than 10–20% to the number of species that they contain.

This brief review emphasizes the diversity of factors that stimulate speciation in certain genera but are apparently lacking in others. Evolution can be stimulated by any one of a great variety of opportunistic interactions between the gene complexes of certain populations and environmental factors that promote their diversification. Given enough data on sizes of genera having different geographical distributions, patterns of morphology and of cytogenetic diversity, one might be able to calculate the frequency with which large genera could be expected to evolve. I do not believe that this is possible with the information we have today.

WHY DO LARGE FAMILIES EVOLVE?

The same kind of problem faces us when we consider different angiosperm families. Eight giants among them are Orchidaceae (800 genera, 30,000 species), Asteraceae (1000 genera, 20,000 spp.), Fabaceae, (sens. lat. or Leguminosae) (700 genera, 10,000 spp.), Poaceae (700 genera, 9000 spp.), Rubiaceae (450 genera, 7000 spp.), Euphorbiaceae (290 genera, 7500 spp.), Apocynaceae (including Asclepiadaceae) (400 genera, 4000 spp.), and Brassicaceae (350 genera, 3000 spp.). A review of these families with respect to the same parameters as those considered in the last section reveals almost the same kind of diversity.

These large families do not differ from the large genera with respect to geographic distribution. None is endemic to a single continent, but the great majority of genera in four of them (Orchidaceae, Rubiaceae, Euphorbiaceae, Apocynaceae) are tropical or subtropical. Only Brassicaceae is predominantly temperate. This difference is in accord with the plausible hypothesis that most plant families evolved and began their diversification during the Cretaceous and early Tertiary period, when frostless climates that are now confined to tropical and subtropical regions were much more widespread (Axelrod 1966).

With respect to age, fossils resembling Fabaceae are known from upper Cretaceous deposits, suggesting that this family is among the oldest of modern plant families (Raven and Axelrod 1974). On the other hand, the Asteraceae are believed to be no older than the early Tertiary period, and perhaps as young as the Upper Oligocene epoch. When one considers the additional fact that one of the smallest angiosperm families, the Platanaceae, is also one of the oldest, while two other giants, Orchidaceae and Poaceae, are also relatively recent, one cannot escape the conclusion that the number of modern species that a family contains is weakly—if at all—correlated with its age or may even be inversely correlated.

If we compare these families with respect to the other parameters considered in the last section, we find the same lack of consistency. Six of them contain a great variety of growth habits, including trees, shrubs, vines, perennial and annual herbs, and succulents. On the other hand, Orchidaceae consist entirely of perennial herbs, most of them epiphytic, whereas Brassicaceae contain only herbs and subshrubs. Six of them include a diversity of pollination mechanisms, but Brassicaceae are relatively stereotyped in this respect, whereas Poaceae are either wind-pollinated or autogamous with monotonous consistency. With respect to mechanisms for seed dispersal, Asteraceae, Fabaceae, Poaceae, Rubiaceae, and Brassicaceae display a rich diversity, but Euphorbiaceae all possess a dry, 1- to 3-seeded capsule, and Apocynaceae and Orchidaceae are monotonously homogeneous, possessing follicles that contain a large number of seeds. Chromosomal diversity is high in Asteraceae, Fabaceae, Poaceae, Euphorbiaceae, and Brassicaceae, but nearly all Apocynaceae have $2n = 22$ with few polyploids, and the majority of Rubiaceae and Orchidaceae resemble each other with respect to karyotype. Families of angiosperms, like genera, may become rich and diverse in species because of any one of a combination of stimulating factors.

WHY SO MANY FAMILIES OF ANGIOSPERMS?

Most angiosperm families evolved between 60 million and 100 million years ago, when environments were very different from those that now prevail. This was particularly true with respect to mammals, birds, and insects, which might have served as pollen and seed vectors. Furthermore, the evidence for differentiation of angiosperm families during the Cretaceous and early Tertiary periods consists almost entirely of fossil leaves and pollen, which to a certain extent make possible the recognition of families but tell us very little about the way in which adaptive characteristics evolved. For the most part, botanists can formulate hypotheses about the reasons for angiosperm diversification at the family level only by discerning what characteristics, either adaptive or adaptively neutral, are most important in differentiating modern angiosperm families from each other.

Botanists who have reviewed the phylogeny of plant families and orders (e.g., Cronquist 1968, Takhtajan 1969, 1980, Thorne 1976) and I (Stebbins 1974) agree that most of the differentiation of families and orders has involved reproductive structures rather than parts of the vegetative system. The distinctive character complexes by means of which systematists recognize orders and families, to the extent that they are adaptive, consist of alternate solutions to problems of crosspollination by various kinds of vectors; seed production and dispersal; and seedling establishment. The last of these processes is intimately connected with the size of the seed and the nature of the stored food that it contains, and so is largely dependent upon genetic changes that affect the gynoecium of the parent plant. Which kinds of adaptations have been the most important in the differentiation of angiosperm families: those concerned with pollination or those related to seed production, dispersal, and seedling establishment?

To answer this question, I considered separately each order of angiosperms as defined by Cronquist (1968, Stebbins 1974, Appendix), to determine in which of them the differentiation of their component families is based more on characteristics of the perianth and androecium than the gynoecium, and in which orders gynoecial differences between families are more important. Disregarding orders that consist of a single family, as well as those in which the component families are all treated differently by other systematists, I reached the following conclusion. In 26 orders, gynoecial characters predominate; in 16 orders, perianth and androecial characters are apparently more significant; and for 9 orders, I could not reach a decision.

The bias in favor of gynoecial characters appears even more significant when one considers their position in the overall phylogenetic scheme. Among the three superorders listed first by Cronquist (Magnoliidae, Hamamelidae, Caryophyllidae), all of the orders that they contain except one (Papaverales) have their component families differentiated from each other more on the basis of gynoecial than androecial characters. On the other hand, the two orders placed in the final superorder, Liliidae, are the Liliales and Orchidales, in both of which family differentiation is based chiefly on perianth and androecial characteristics. The same is true of the Scrophulariales and Campanulales in the advanced superorder Asteridae, in which pollination is chiefly carried out by the most specialized of insect vectors in the orders Hymenoptera, Diptera, and Lepidoptera.

I conclude, therefore, that most of the differentiation of families during early stages of angiosperm evolution was based upon characters affecting the gynoecium. These are to a large extent different adaptive strategies for securing maximum seed production, seed dispersal, and competitive survival of seedlings under various environments. Evolution of elaborate strategies for crosspollination by insects and other pollen vectors took place mostly during later stages of the evolutionary differentiation of angiosperm families and genera. By the time strategies for achieving crosspollination became a principal theme of their evolution, angiosperms had already become the dominant plants of the earth's vegetation. The multiplication of their species was by that time a foregone conclusion.

EARLY ANGIOSPERM CHARACTERISTICS PROMOTING DIVERSIFICATION

Crosspollination by insect and other pollen vectors has played an important role in generating the current large number of species of flowering plants. But this factor has not been the dominant one except during the more recent phases of angiosperm evolution, when the great majority of the 250 families had already evolved and become stabilized and when many modern genera existed. What characteristics, then, gave the angiosperms their tremendous advantage over other vascular plants, and enabled them to replace ferns, conifers, and other gymnosperms during the middle of the Cretaceous period (Doyle and Hickey 1976)?

I believe that the key role must be ascribed to changes that affected seed production, dispersal, and seedling establishment. Paramount among them were the cytological and developmental changes that made possible a drastic shortening of the time required from the initial differentiation of ovules through anthesis and pollination, up to the growth of the embryo and ripening of seeds. Perhaps the first of these was the reduction of the female gametophyte to an eight-nucleate embryo sac that requires for its completion only three mitotic divisions and no cell wall formation. Second was the evolution of double fertilization, giving rise to triploid endosperm. This acquisition made possible rapid development of endosperm, which, in turn, promotes rapid early growth of the embryo. Third was the closed carpel, providing maximum protection already at anthesis for tender growing ovules and reducing the adaptive necessity for massive integuments. Fourth was the stylar canal, which promotes rapid growth of pollen tubes and greatly reduces the amount of time required between pollination and fertilization.

All these evolutionary innovations equipped early angiosperms with a maximum amount of reproductive flexibility or opportunism. Evolution of flowers capable of producing many small seeds became possible but not necessary. Angiosperms could evolve the ability to colonize not only pioneer habitats in semi-arid regions, in which rapid production and wide dispersal of seeds has a high adaptive value (Stebbins 1974), but also could evolve climax trees in rainforests. There, the equable climate permits a long continued reproductive cycle, and large seeds are, as a rule, advantageous because of the large amount of stored food that they contain. This enables the seedlings to grow vigorously in the dimly lighted forest floor, since they do not depend upon photosynthesis until they have reached a considerable size. A notable feature of many rainforest trees belonging to diverse families (Lauraceae, Burseraceae, Mimodoideae, Caesalpinoideae, Dipterocarpaceae) is that they have small flowers but relatively large fruits and seeds. In some of these families, food is stored in the cotyledons of the embryo rather than as endosperm, making it particularly accessible to the growing plant.

The more specialized anatomy of their stems has contributed greatly to the flexibility and success of the angiosperms. Although a few vesselless angiosperms are well known, they are rare and probably played a minor role in the expansion and success of the class as a whole. The differentiation of vessels and wood fibers played a major role in the conquest by angiosperms of arid regions, in the ability to evolve efficient species having deciduous leaves (Carlquist 1975), and probably also in the evolution of rapidly growing herbs.

Genetic flexibility with respect to the reproductive cycle was not the only kind of flexibility that promoted the early diversification of angiosperms. Recent fossil evidence indicates that even the earliest known angiosperms included a diverse array of growth habits. Vakhrameev and Krassilov (1979) have described a stem and inflorescence (*Caspiocarpus*) from central Asiatic strata of early Cretaceous (Albian) age that almost certainly belonged to an herbaceous plant. It is associated with leaves similar to those long known from the Potomac series of the eastern United States, described as *Vitiphyllum*, which because of its resemblance to *Caspiocarpus* might also belong to an herbaceous form. Most systematists concerned with angiosperm phylogeny (Cronquist 1968, Thorne 1976) regard the Nymphaeales and Nelumbonales as primitive and probably ancient. Doyle and Hickey (1976) have suggested that the early Cretaceous leaves from the Potomac formation described as *Acaciaephyllum* may have belonged to an herbaceous monocotyledon. All of these early forms existed when angiosperms were comparatively rare and when they probably included a relatively small number of species, before their rise to prominence. During the period of great expansion for the class, the upper Cretaceous (Cenomanian), different groups of angiosperms having a variety of sizes and growth forms may have been evolving simultaneously. This probable situation is not clearly shown by the fossil record, which is strongly biased in favor of those kinds of plants that are most easily preserved: woody species living near streams and lakes.

In addition to reproductive flexibility, did the earliest angiosperms possess any other characteristics that favored the evolution of a diversity of growth habits? A plausible answer to this question cannot be given until much more is known about the comparative physiology of angiosperms and other seed plants. Without being able to find any hard data in support of these notions, I have gained the impression that in angiosperms two developmental characteristics are much more prominent than they are in the various orders and families of gymnosperms. One is the capacity of their cells to elongate in response to hormonal stimulation. This effect is particularly noticeable in the much longer root hairs of many angiosperms as compared to conifers; in the frequent presence in them of elongate trichomes, consisting of one or a few cells; in the frequent production of tendrils and prehensible leaf stalks in angiospermous vines; and in the etiolated elongation of stems that is a response to low light. Are these tendencies due to the greater efficiency with which growth substances such as indole acetic acid and gibberellins are synthesized in angiosperms as compared to other seed plants or to chemical features of their cell wall structure that make angiosperm cells more sensitive to the action of these substances, or to both properties? If most modern angiosperms are distinctive in these respects, did these distinctive characteristics evolve early or relatively late in the evolutionary history of the class? Clearly, the growth of the now almost nonexistent discipline of comparative developmental physiology will do much to answer basic questions about angiosperm evolution.

Another feature of angiosperm development that is much better developed than in other vascular plants—and is particularly conspicuous in the more advanced monocotyledons—is the activity of intercalary meristems. Althugh subapical meristematic growth is well developed in the developing twigs and leaves of many conifers, no conifer can match the situation found in most genera of grasses, such as barley and wheat, in which the entire flowering culm, including leaves, spikelets, and florets, becomes differentiated before it grows out of the sheaths of the surrounding leaves. Moreover, the principal advances in floral specialization that characterize the more advanced orders and families of angiosperms, particularly sympetaly and epigyny, evolved via a transfer of growth from apical to intercalary meristems (Stebbins 1974, chap. 6). The evolution of intercalary meristems that are active at particular developmental stages and for defined periods of time must require a system of regulatory genes that is far more complex than those in other vascular plants. Here, also, no evidence is available concerning whether increased complexity of gene regulation existed already among the earliest angiosperms or whether its evolution was a byproduct of evolution in other directions.

CONCLUSIONS

Present knowledge permits no simple answer to the question of why there are so many species of angiosperms. Several factors may be involved: First, greater flexibility with respect to seed production, dispersal, and seedling establishment. This was made possible by reduction of the female gametophyte, double fertilization, closure of carpels, union of carpels to form a compound ovary, development of a stylar canal, and (in many families) storage of food in cotyledons rather than endosperm. Second, greater genetic and phenotypic flexibility with respect to cell and shoot elongation, because of more efficient synthesis of growth substances or more sensitive responses of cells to these substances. Third, more complex mechanisms for activating and repressing the activity of genes, reflected in the development of more extensive and more complex intercalary meristems. Fourth, greater complexity in the flower, permitting in many evolutionary lines the shift from insect to wind pollination and vice versa, diversification of flowers as adaptations to the visits of various kinds of vectors, and (in some groups) evolution of many variants upon a particularly complex theme of flower-pollinator interaction. I have listed these factors in order of their importance, as I see it, on the basis of current data. Nevertheless, many relevant data that might radically change the evaluation remain to be obtained.

The principal themes of angiosperm evolution are still imperfectly known. New and relevant data can be expected from many directions, including reproductive biology of seeds and seedlings, pollination biology, developmental genetics and physiology, and paleobotany. For evolutionary botanists, the coming half-century is likely to be as exciting and as full of surprises as the last.

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Why So Many Kinds . . .

Continued from p. 572

cambrian-Cambrian boundary, followed by an equilibrium of family diversity lasting until a second period of diversification tripled the number of families in the Ordovician, 100 million years later. Sepkoski interpreted this as an early success of generalists, later replaced by more specialized forms attaining a higher diversity level that lasted until the great Permo-Triassic marine extinctions.

The angiosperms, I believe, have made possible a similar rise to a new and much higher level of terrestrial species diversity, both by developing species-rich vegetation with a great many co-dominant trees and fast turnover rates and by exploiting previously sparsely populated habitats now covered by grasslands, thornbush, and alpine meadows. Bambach (1977) suggested that one explanation of sudden increases in marine within-community species richness in the Silurian and in the Cretaceous-Cenozoic transition may have been improved nutrient supplies in the oceans following changes in the terrestrial flora—the first following the Silurian spread of early land plants and later following the Cretaceous expansion of the angiosperms (cf. Knoll et al. 1979, Niklas et al. 1980).

ANGIOSPERM SUCCESS

Recent reviews of the evolution of terrestrial plant diversity (Whittaker 1977), tropical plant diversity (Baker 1970), and insect diversity (Lawton 1978, May 1978, Southwood 1978) have not emphasized the point I am making here—namely, that the angiosperms are largely responsible for today's terrestrial species richness. Characteristics that have allowed angiosperms to outcompete and outnumber spore-bearing plants and gymnosperms include a rapid reproductive cycle, flexible modes of breeding behavior, improved protection and dispersal capacity for seeds, extraordinary variety in growth form and anatomy, diverse systems of chromosome repatterning, and varied arrays of biochemical defenses. Pollen tube competition, allowing natural selection to operate directly on the haploid phase of the life cycle, may have been another factor in angiosperm success (Mulcahy 1979). Angiosperm leaf evolution also played a significant role; two morphological innovations were especially important in the adaptation to strongly seasonal environments. The most significant of these was the strong, but thin, reticulately veined leaf that characterizes the deciduous dicotyledonous tree. The compound leaf with "deciduous twigs" is an extension of this development and plays an important role in certian hot and dry environments (Givnish 1978). The second important innovation was leaf growth from basal meristems protected in a tightly congested tuft, as in many grasses, which allowed these plants to become a dominant form on broad open plains where desiccation, intense fire, and heavy grazing are frequent.

Although these factors all played a role in the achievement of angiosperm dominance, the most important innovation may have been pollination by insects, which allows outcrossing sexual reproduction in highly dispersed populations of relatively few individuals (Müller 1877). The ability to maintain gene exchange at low population densities together with a faster reproductive cycle and animal-dispersed propagules probably reduced the extinction rate of angiosperms in the face of changing environments and predation (Raven 1977, Regal 1977).

Once flowering plants had established themselves and after their initial diversification, the process of active speciation in many parallel lineages greatly enlarged the terrestrial resource spectrum. As an opposing force, progressive evolution, constantly selecting for more effective competitors, should have enhanced competitive exclusion and thus reduced or equilibrated species richness over time. But many factors, especially niche-partitioning, have counteracted competitive exclusion. Moreover, processes of rarefaction, such as predation, disease, and unpredictable disruption, have counteracted the expansion of superior competitors (Connell 1978, Huston 1979, Levin 1974).

As communities have become richer in species and more complex in their interactions, replacement success may have become more subject to stochastic effects. The success of a tropical forest tree may be due more to the chance of having its seedling or sapling in the right place at the right time than to its special competitive attributes (Burger 1980, Hartshorn 1978). In rich tropical communities survival at, or recovery from, very low population levels is especially important, and it is in these very rich communities where the flowering plants have been most successful in replacing the gymnosperms.

Gymnosperm forests provide a striking contrast to angiosperm-dominated forests. For example, species richness occasionally decreases on environmental gradients where one would expect richness to increase, as from tall moist forest to drier open woodland or in the transition from cool montane forest to colder alpine scrub. These diversity depressions are usually associated with gymnosperm dominance, as in the coastal redwood to interior woodland transition in northern California or in the montane conifer to alpine meadow transition (Glenn-Lewin 1977, Peet 1978, Slack 1977, Whittaker 1977).

The diversity depression associated with gymnosperm dominance, whatever its functional cause, lends credence to the notion that an entire world dominated by these plants was a world much poorer in species. The exclusion of today's majestic, but species-poor, conifer forests from tropical moist lowlands and mid-elevations may be a reflection of their inability to compete in rapid turnover, species-rich vegetation. Tall forests, dominated by many large old trees